Power

File: power.kicad_sch

MCU

File: mcu.kicad_sch

Driver 1 and matrix

File: matrix1.kicad_sch

Driver 2 and matrix

File: matrix2.kicad_sch

Driver 3 and matrix

File: matrix3.kicad_sch

Driver 4 and matrix

File: matrix4.kicad_sch

Ultra-dense 20x20 mm square 0808 LED array (s15-225): 225 MHPA0808RGBDT 0808 LEDs (15x15 square in 20x20 mm, 1.36 mm pitch, 56.3 LEDs/cm^2), 4x AW20198QNR, 8-layer ENIG.
25 x 50 mm rectangular PCB: North 20x20 mm window holds ONLY the dense MHPA0808RGBDT LED matrix;
all drivers, MCU, flash, charger, and sensors sit in the South electronics bay (0 via-in-pad).

Drawn by scripts/dense_square/sch_gen.py

225 MHPA0808RGBDT 0808 LEDs (15x15 square in 20x20 mm, 1.36 mm pitch, 56.3 LEDs/cm^2), 4x A

CERN-OHL-P-2.0

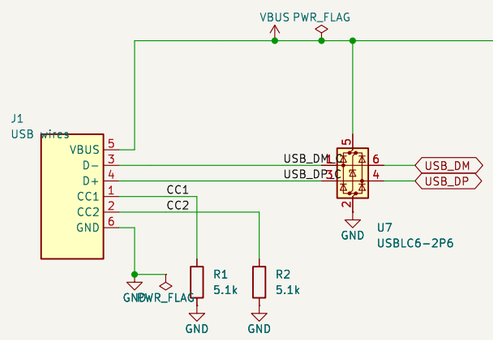
Sheet: /

File: dense-square.kicad_sch

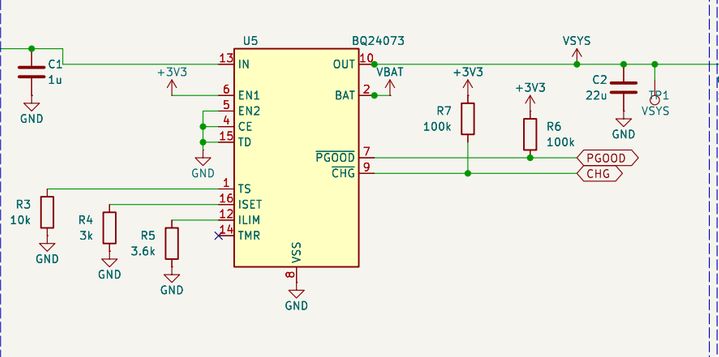
Title: dense-square (s15-225): top

Size: A4	Date: 2026-11-06	**Rev: S15**
KiCad E.D.A. 9.0.8		Id: 1/7

USB WIRES, ESD

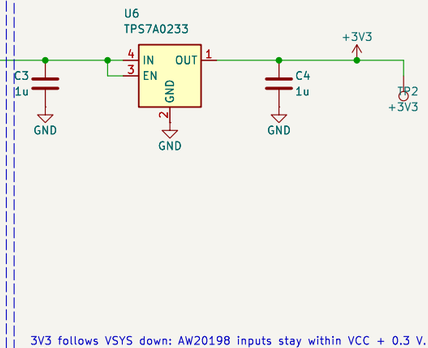


CHARGER AND POWER PATH



USB500: 450–500 mA in, 297 mA charge, ILIM must not float.
TMR open: default 5 h. TS fixed: no NTC.

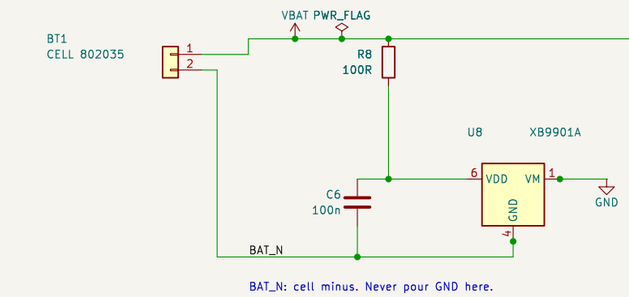
3V3



3V3 follows VSYS down: AW20198 inputs stay within VCC + 0.3 V.
150 mA max.

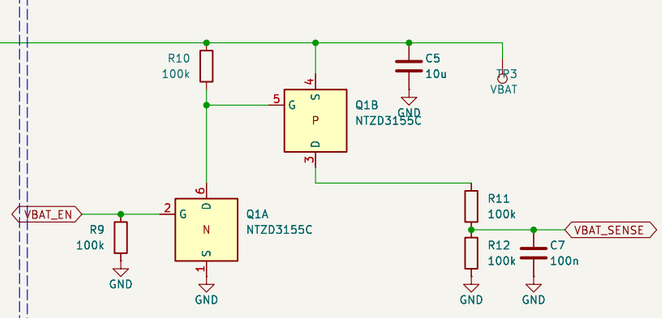
CELL AND PROTECTION

Hand-soldered leads. Standard 802035 / 102035 (20x35 mm) prismatic LiPo cell.



BAT_N: cell minus. Never pour GND here.

VBAT SENSE



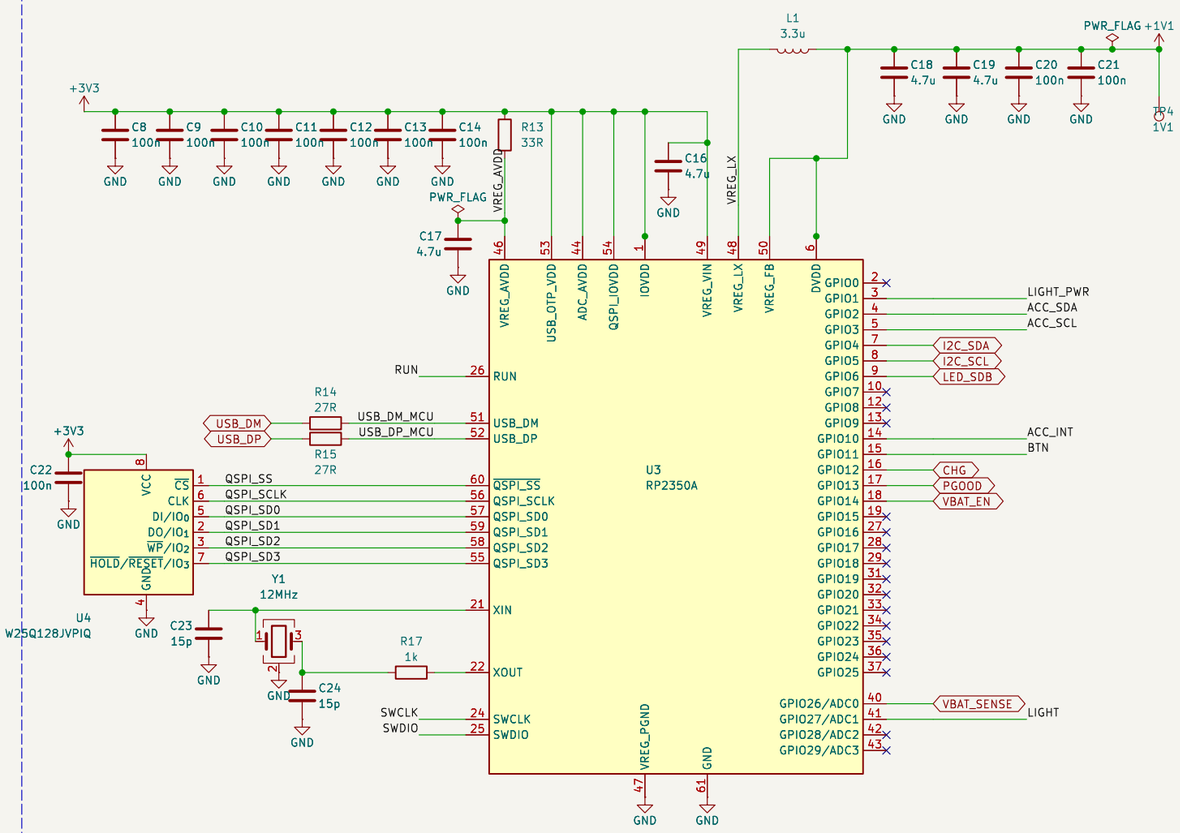
Drawn by scripts/dense_square/sch_gen.py
USB-C wires, charger, protection, LDO, VBAT sense (prismatic 802035 cell)
CERN-OHL-P-2.0

Sheet: /Power/
File: power.kicad_sch

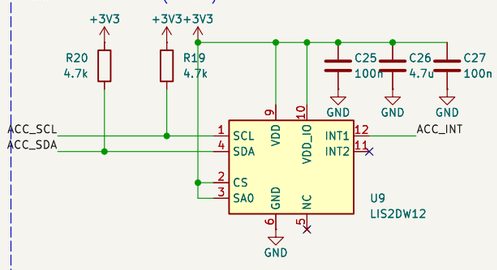
Title: dense-square (s15-225): power

Size: A3	Date: 2026-11-06	**Rev: S15**
KiCad E.D.A. 9.0.8		Id: 2/7

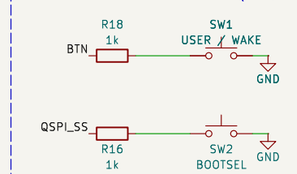
RP2350A, FLASH, CRYSTAL



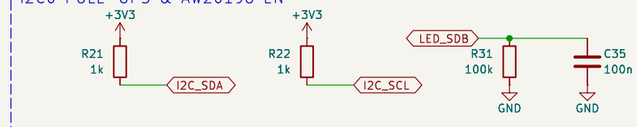
ACCELEROMETER (WAKE)



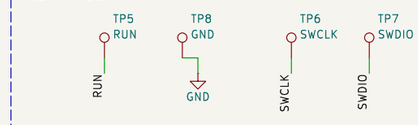
PEG-LESS SIDE BUTTONS (EVQP7J01P)



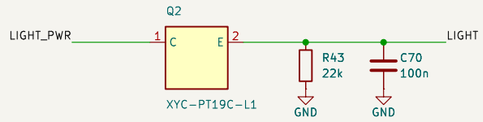
I2C0 PULL-UPS & AW20198 EN



TEST PADS



AMBIENT LIGHT SENSOR



Drawn by scripts/dense_square/sch_gen.py

RP2350A, flash, crystal, accelerometer, peg-less side buttons, South-bay light sensor

CERN-OHL-P-2.0

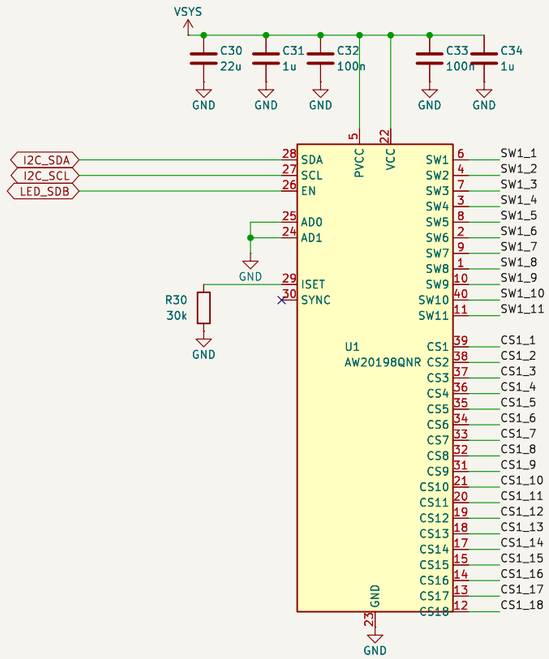
Sheet: /MCU/

File: mcu.kicad_sch

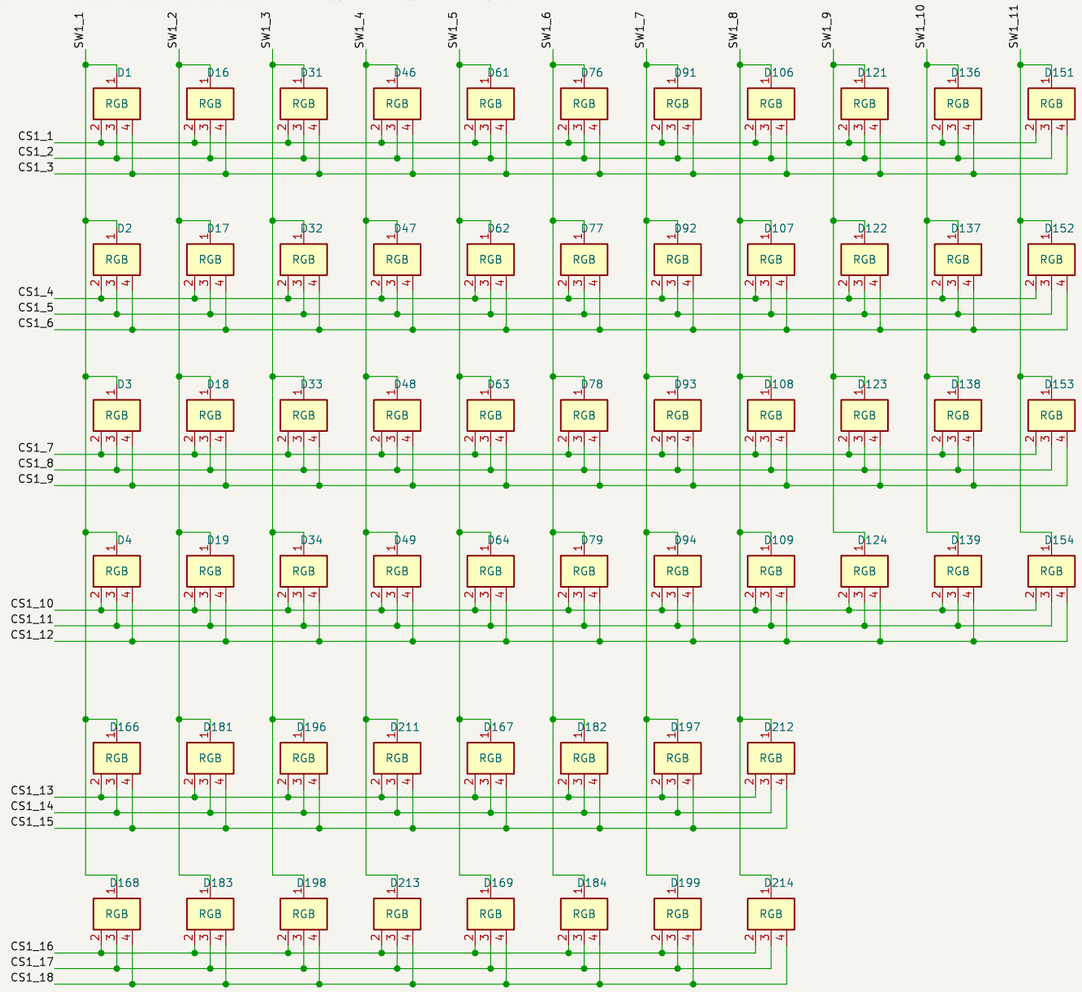
Title: dense-square (s15-225): mcu

Size: A3	Date: 2026-11-06	**Rev: S15**
KiCad E.D.A. 9.0.8		Id: 3/7

DRIVER 1 (U1, 0x20)



60 MHPA0808RGBDT 0808 LEDS: A=Pin1, R=Pin2, G=Pin3, B=Pin4



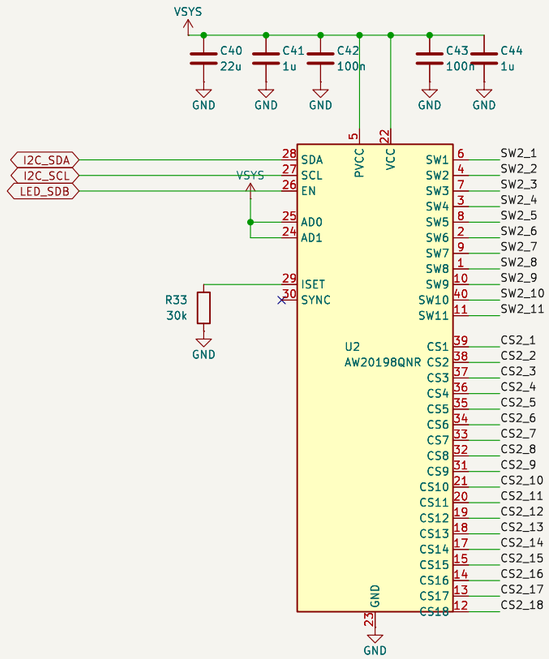
Drawn by scripts/dense_square/sch_gen.py
U1 AW20198QNR (0x20), 60 MHPA0808RGBDT 0808 LEDs (leds-mhpa0808.md)
CERN-OHL-P-2.0

Sheet: /Driver 1 and matrix/
File: matrix1.kicad_sch

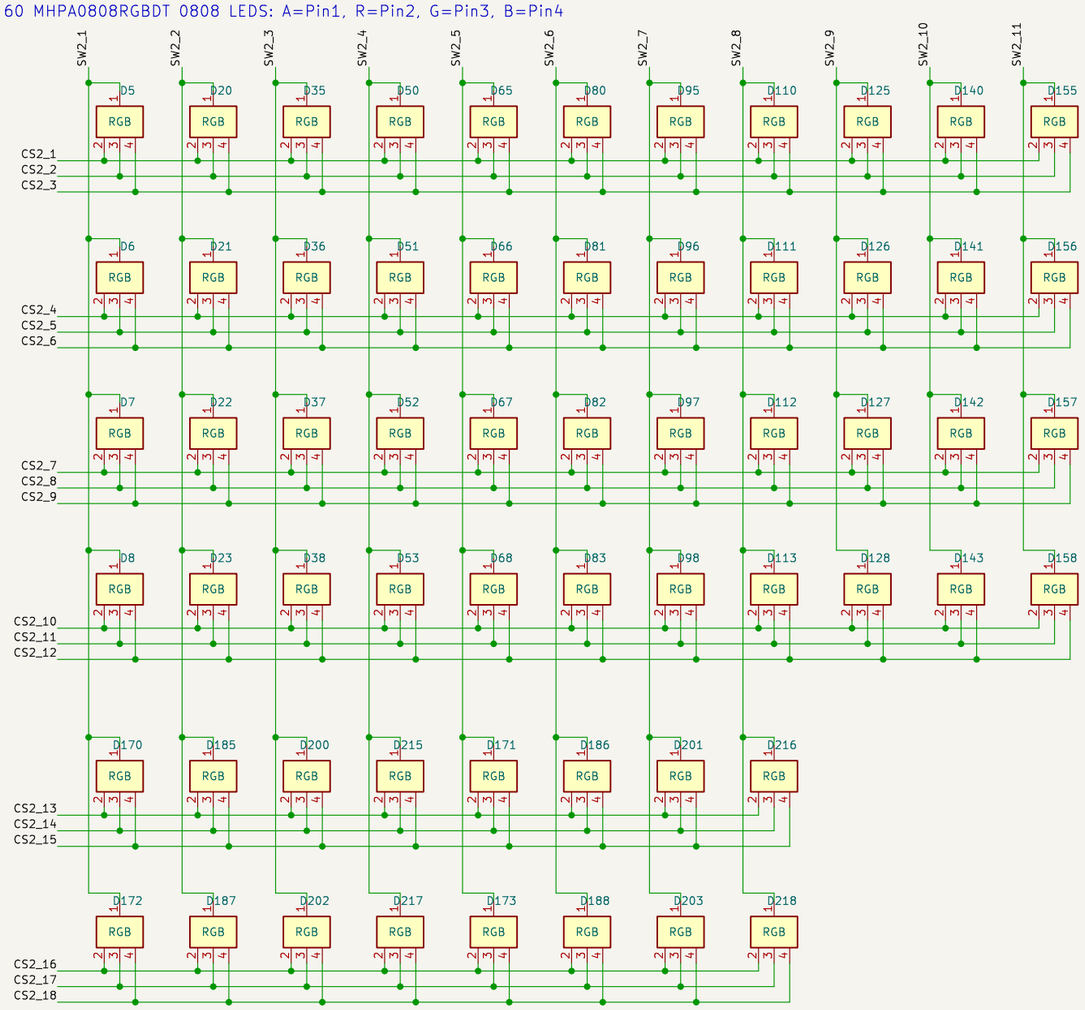
Title: dense-square (s15-225): driver 1 and matrix

Size: A3	Date: 2026-11-06	**Rev: S15**
KiCad E.D.A. 9.0.8		Id: 4/7

DRIVER 2 (U2, 0x25)



60 MHPA0808RGBDT 0808 LEDS: A=Pin1, R=Pin2, G=Pin3, B=Pin4



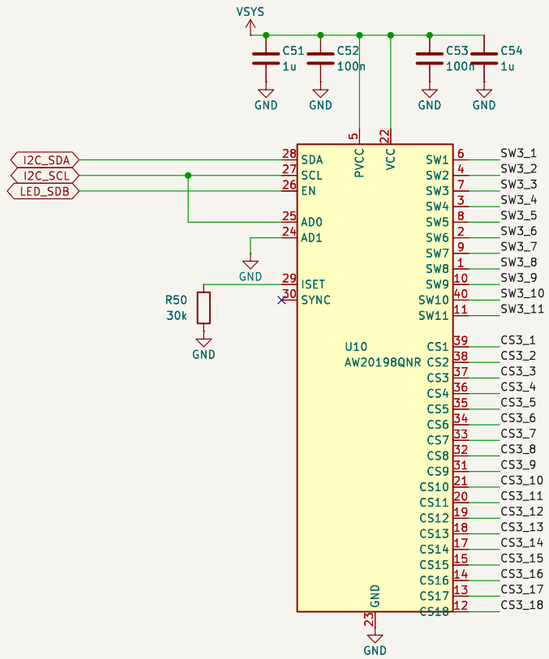
Drawn by scripts/dense_square/sch_gen.py
U2 AW20198QNR (0x25), 60 MHPA0808RGBDT 0808 LEDs (leds-mhpa0808.md)
CERN-OHL-P-2.0

Sheet: /Driver 2 and matrix/
File: matrix2.kicad_sch

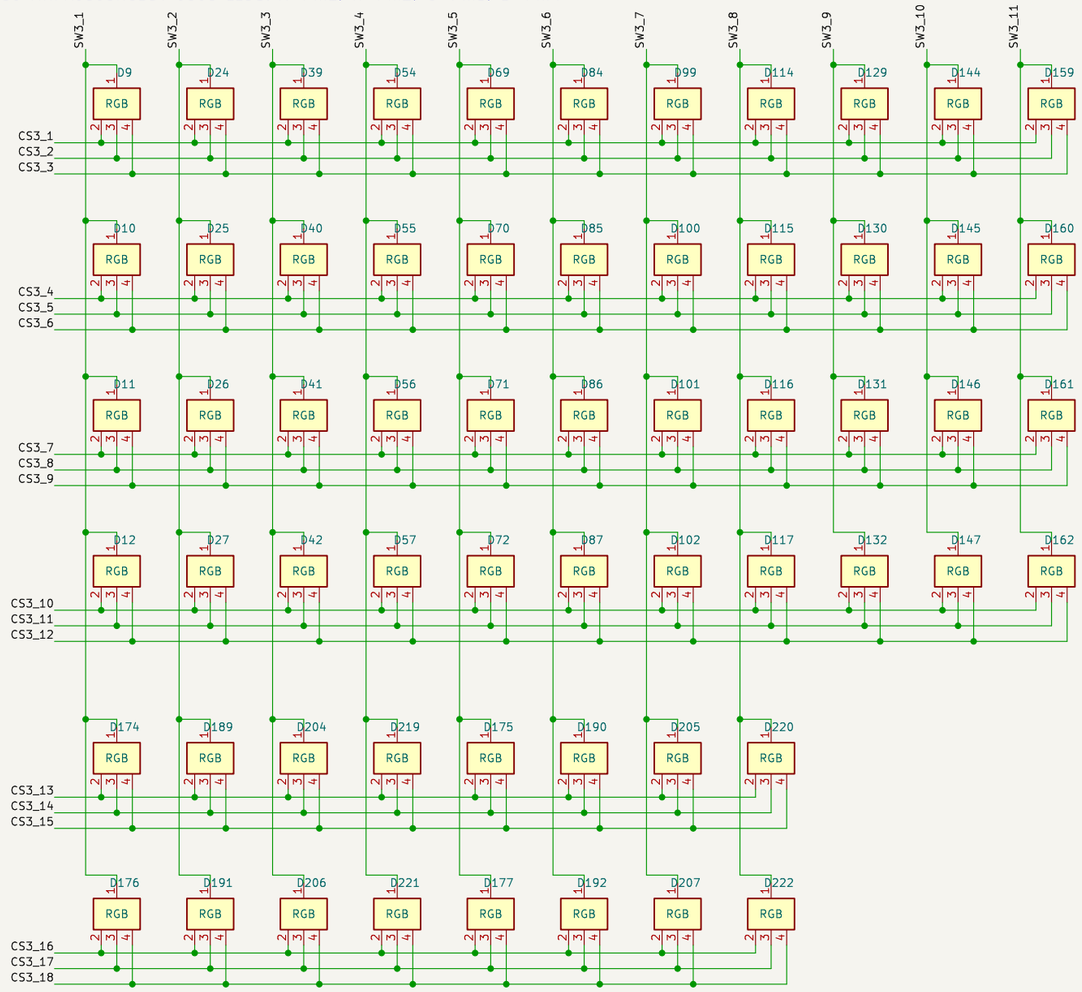
Title: dense-square (s15-225): driver 2 and matrix

Size: A3	Date: 2026-11-06	**Rev: S15**
KiCad E.D.A. 9.0.8		Id: 5/7

DRIVER 3 (U10, 0x22)



60 MHPA0808RGBDT 0808 LEDS: A=Pin1, R=Pin2, G=Pin3, B=Pin4



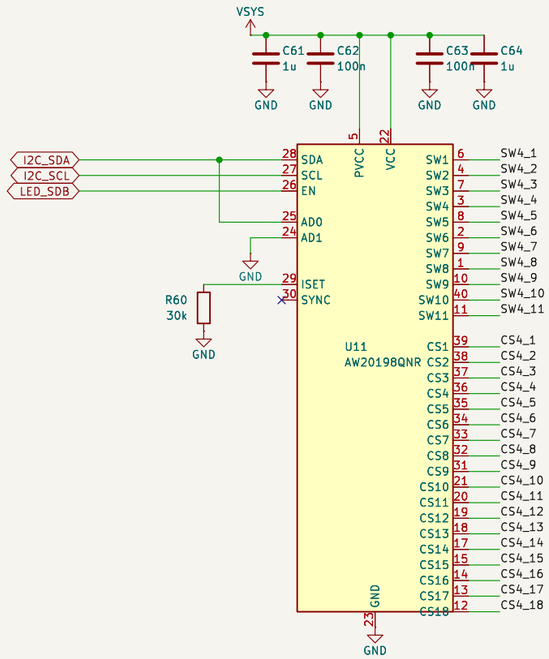
Drawn by scripts/dense_square/sch_gen.py
U10 AW20198QNR (0x22), 60 MHPA0808RGBDT 0808 LEDs (leds-mhpa0808.md)
CERN-OHL-P-2.0

Sheet: /Driver 3 and matrix/
File: matrix3.kicad_sch

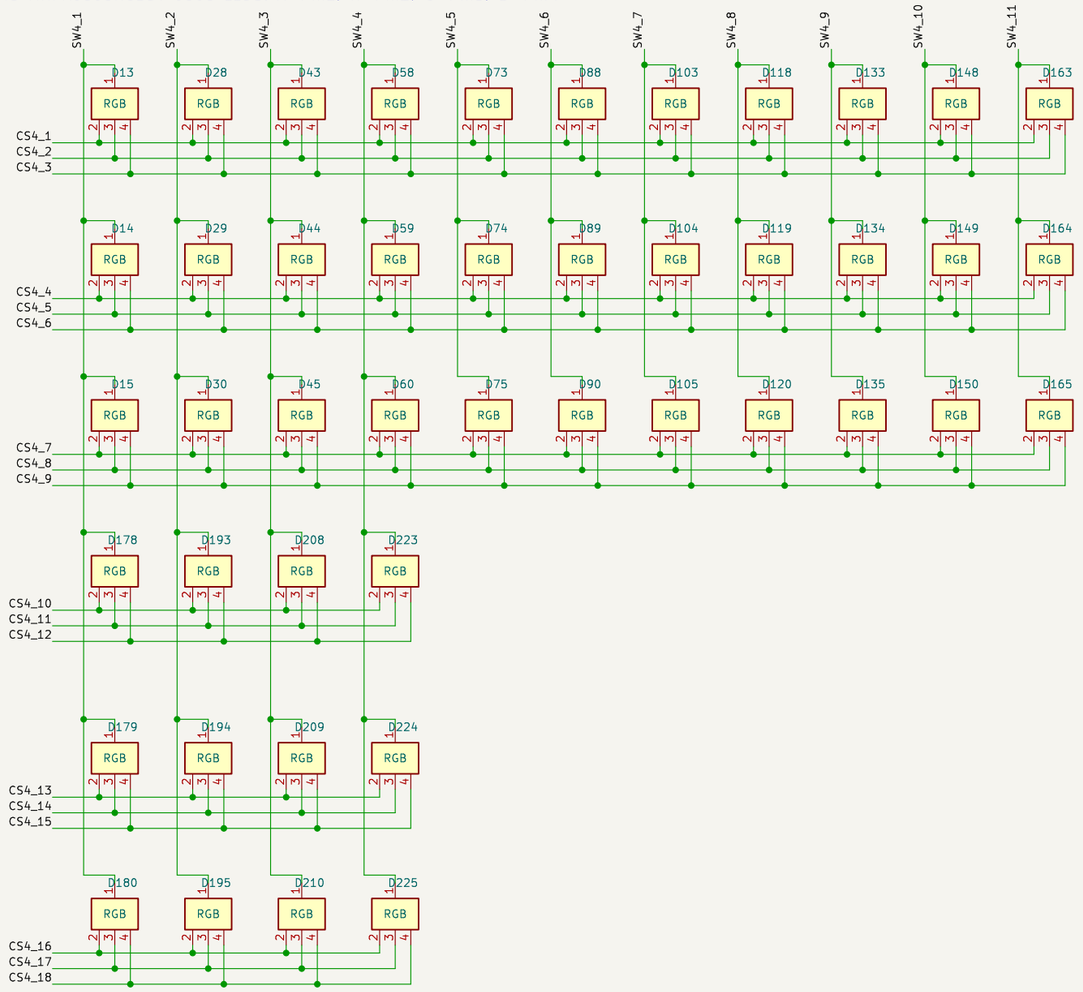
Title: dense-square (s15-225): driver 3 and matrix

Size: A3	Date: 2026-11-06	**Rev: S15**
KiCad E.D.A. 9.0.8		Id: 6/7

DRIVER 4 (U11, 0x23)



45 MHPA0808RGBDT 0808 LEDS: A=Pin1, R=Pin2, G=Pin3, B=Pin4



Drawn by scripts/dense_square/sch_gen.py
U11 AW20198QNR (0x23), 45 MHPA0808RGBDT 0808 LEDs (leds-mhpa0808.md)
CERN-OHL-P-2.0

Sheet: /Driver 4 and matrix/
File: matrix4.kicad_sch

Title: dense-square (s15-225): driver 4 and matrix

Size: A3	Date: 2026-11-06	**Rev: S15**
KiCad E.D.A. 9.0.8		Id: 7/7